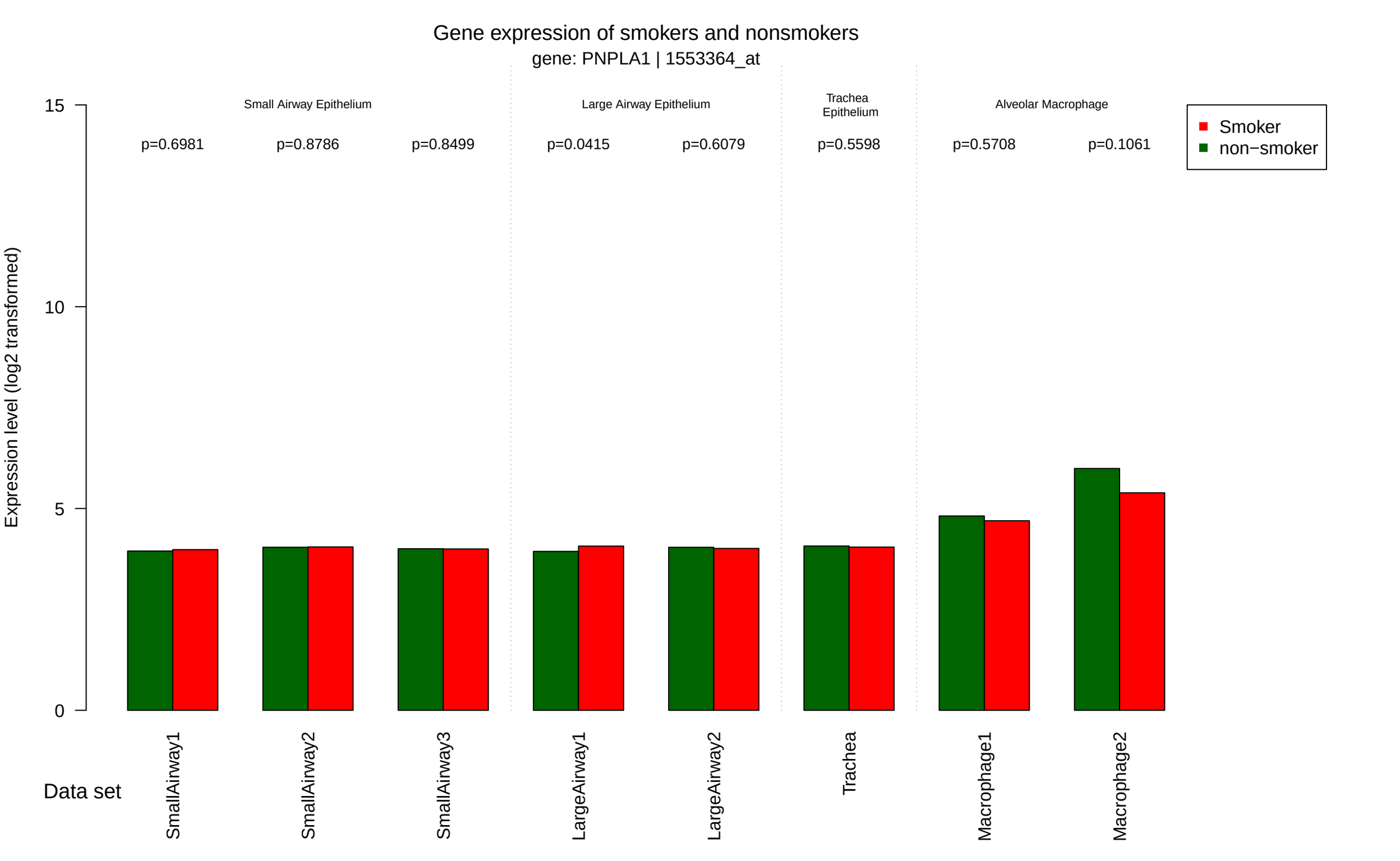
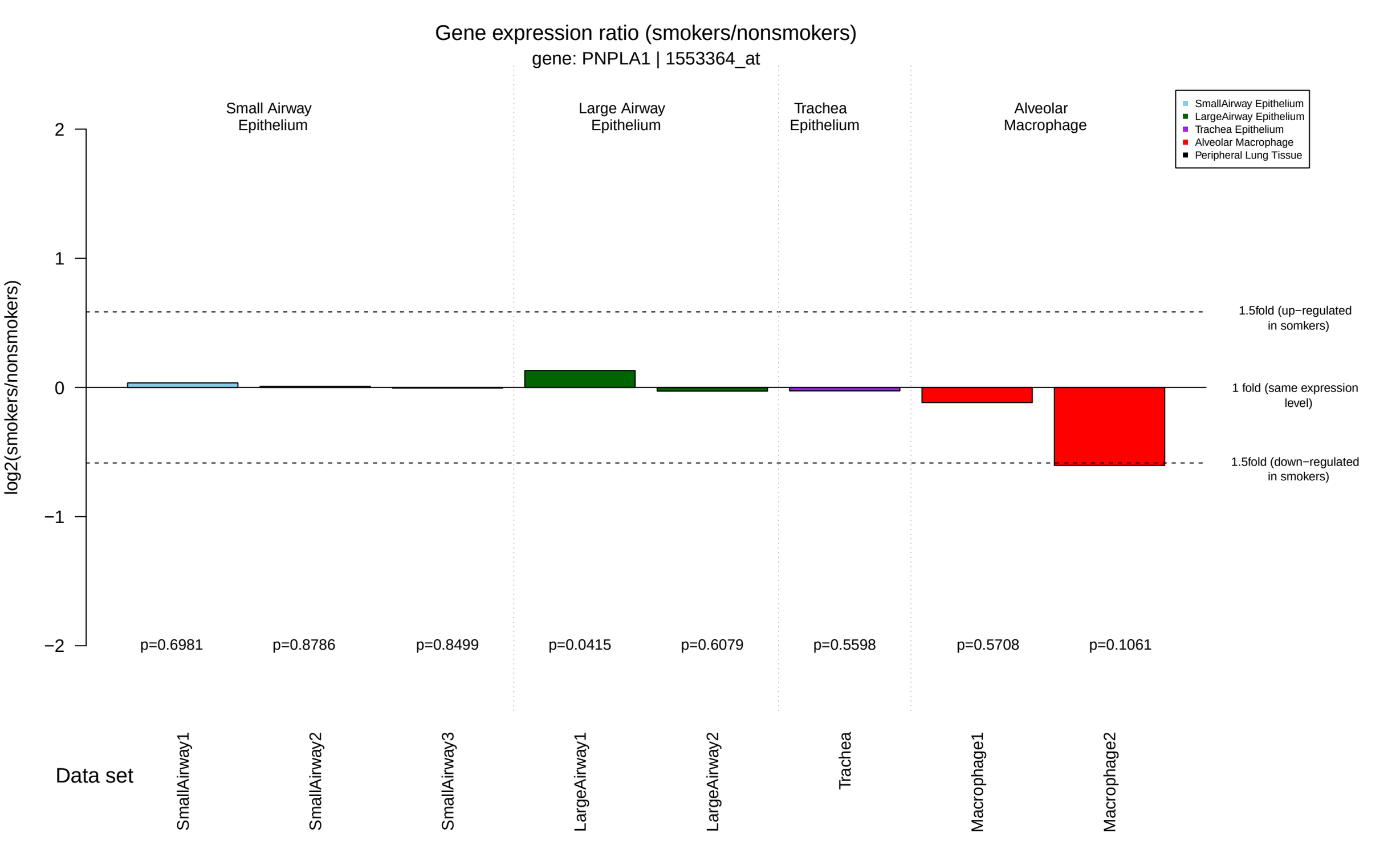
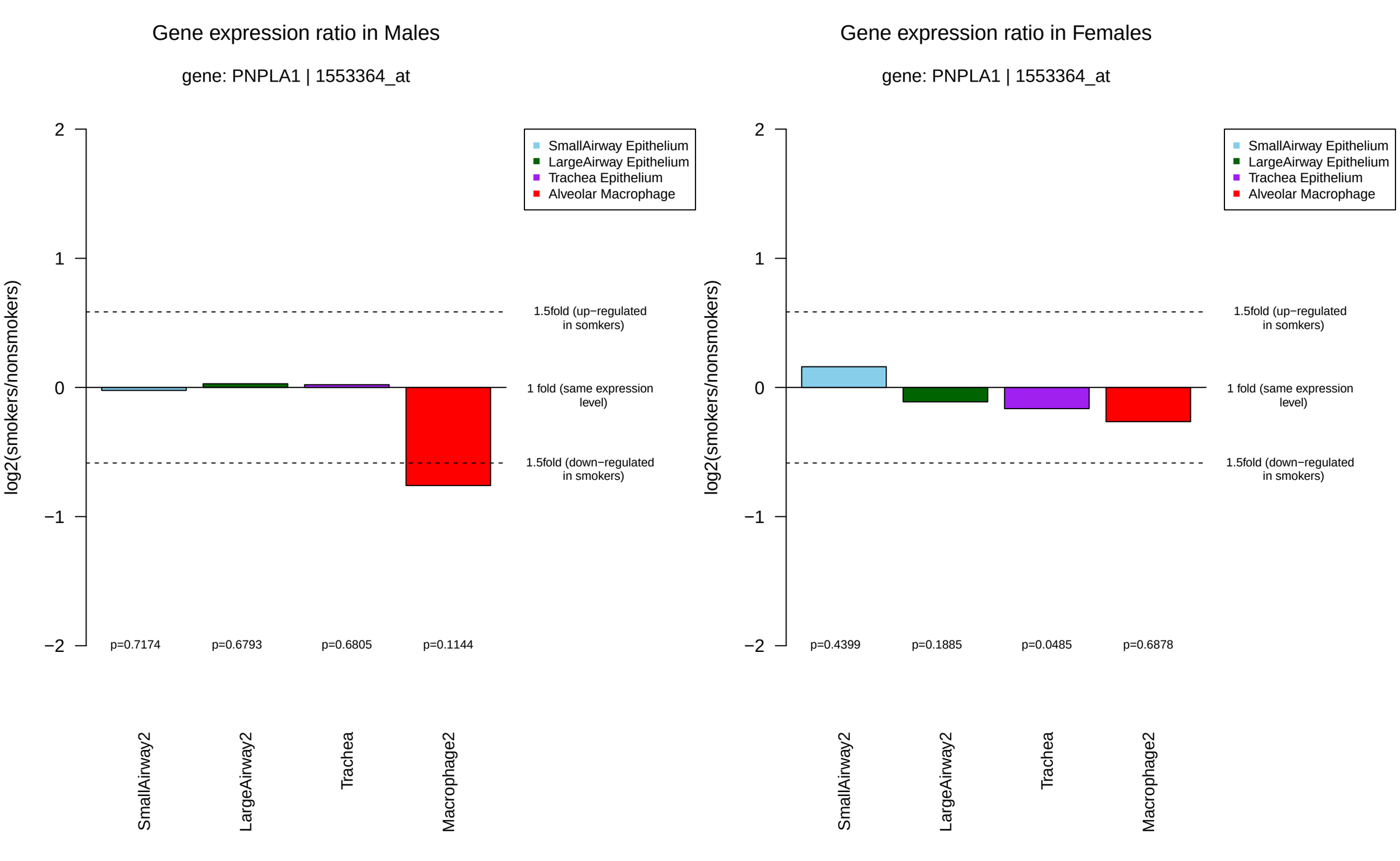
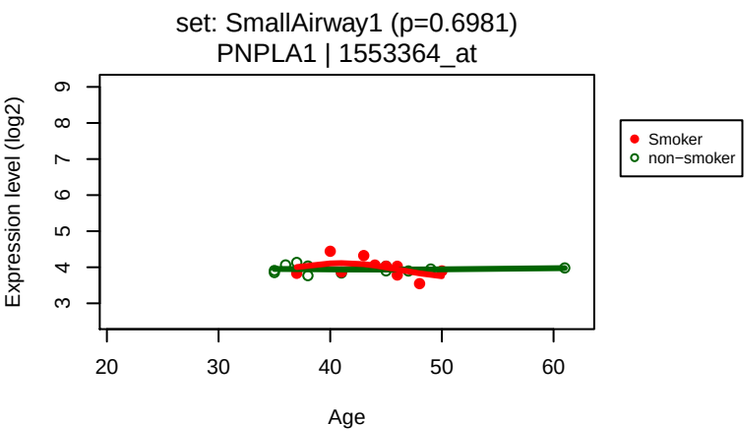
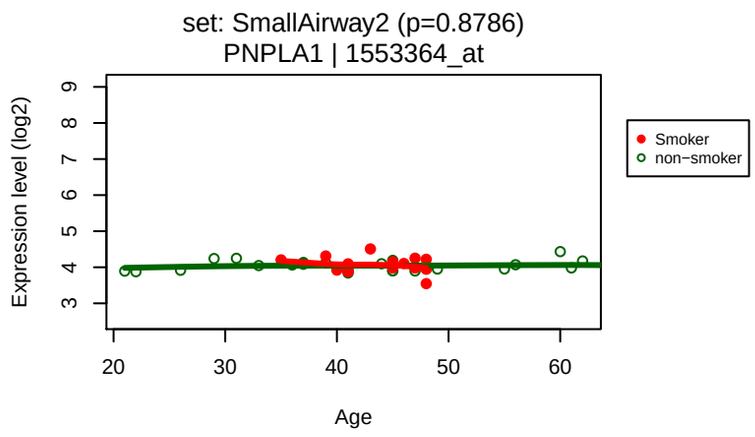
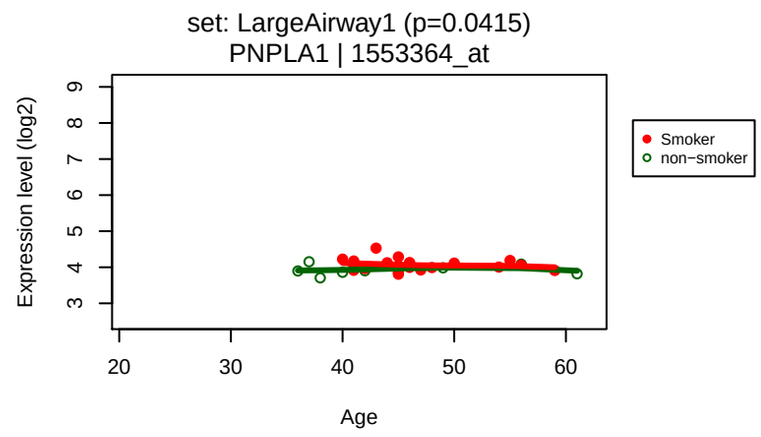

Gene expression of smokers and nonsmokers
gene: PNPLA1 | 1553364_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.6981
p=0.8786
p=0.8499
p=0.0415
p=0.6079
p=0.5598
p=0.5708
p=0.1061
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

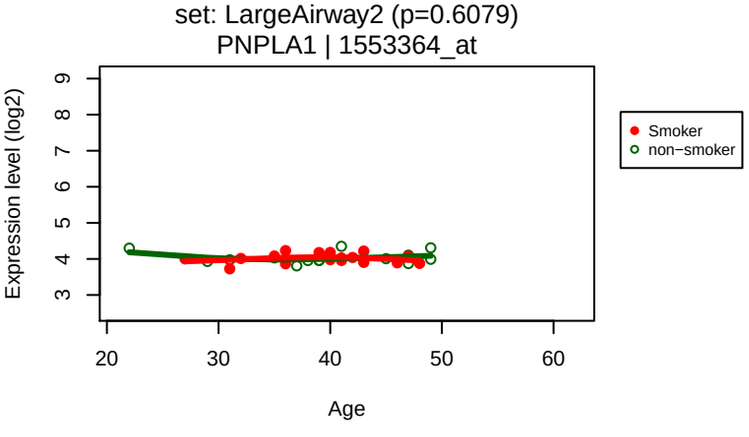

Gene expression ratio (smokers/nonsmokers)
gene: PNPLA1 | 1553364_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.6981
p=0.8786
p=0.8499
p=0.0415
p=0.6079
p=0.5598
p=0.5708
p=0.1061
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

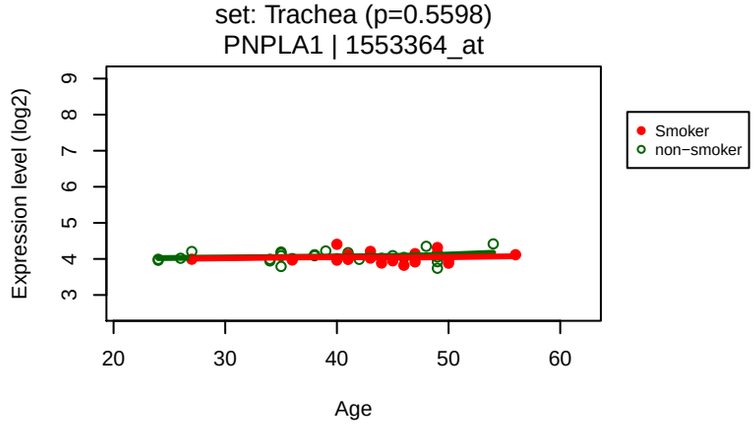

Gene expression ratio in Males
gene: PNPLA1 | 1553364_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.7174
p=0.6793
p=0.6805
p=0.1144
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: PNPLA1 | 1553364_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.4399
p=0.1885
p=0.0485
p=0.6878
SmallAirway2
LargeAirway2
Trachea
Macrophage2

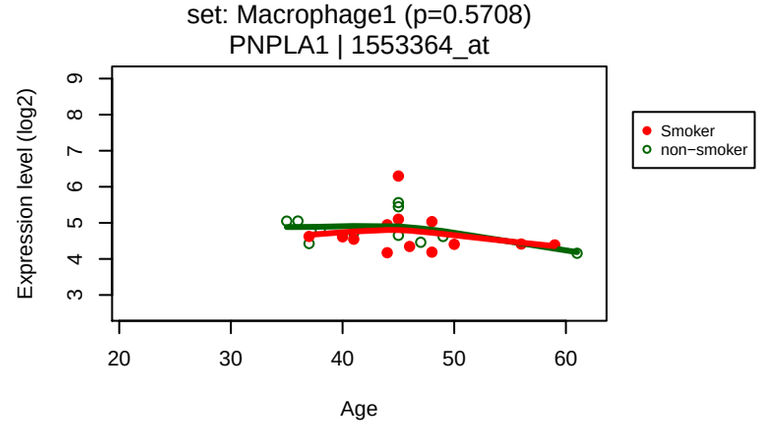

set: SmallAirway1 (p=0.6981)
PNPLA1 | 1553364_at
Expression level (log2)
Age
Smoker
non−smoker

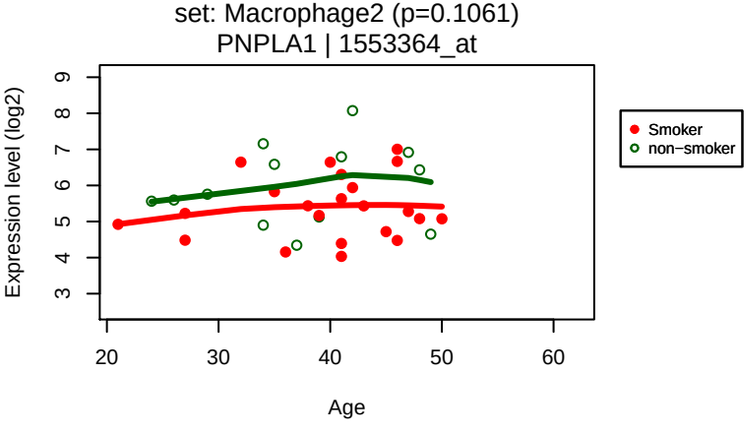

set: SmallAirway2 (p=0.8786)
PNPLA1 | 1553364_at
Expression level (log2)
Age
Smoker
non−smoker

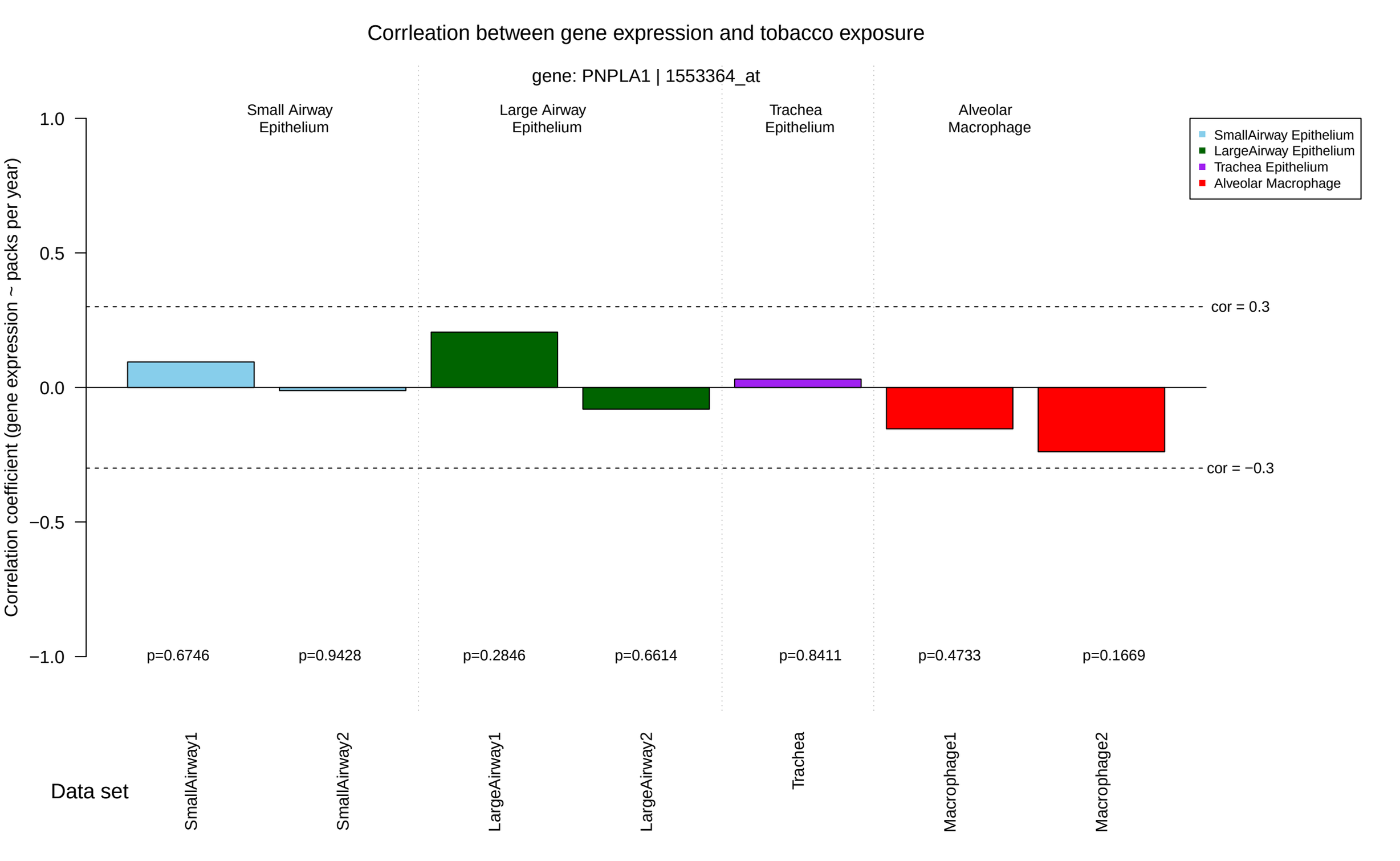

set: LargeAirway1 (p=0.0415)
PNPLA1 | 1553364_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.6079)
PNPLA1 | 1553364_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.5598)
PNPLA1 | 1553364_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.5708)
PNPLA1 | 1553364_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.1061)
PNPLA1 | 1553364_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: PNPLA1 | 1553364_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.6746
p=0.9428
p=0.2846
p=0.6614
p=0.8411
p=0.4733
p=0.1669
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2